GPU Waste Detection for Kubernetes: Market Research, Competition & Unit Economics

Scope: the product **as it is actually built today** (September 2026), not the broader vision. That product is a read-only, self-hosted Helm install that reads NVIDIA DCGM telemetry plus Kubernetes pod identity, classifies GPU waste into four categories (idle, over-provisioned, fragmented, recurring pattern), prices each finding in INR against a right-sizing ceiling, and shows it on a dashboard. It recommends; it does not act. It has been validated on GKE (L4), on a vast.ai/k3s GPU node, against the real Microsoft Philly training trace (23.0% of GPU-workload-hours flagged as waste) and the real Azure LLM inference trace (19,366/19,366 requests served, over-provisioning finding fired).

How to read the numbers: every figure is tagged **[S]** (sourced, see Sources at the end) or **[A]** (our assumption, listed in the Assumptions ledger). Market-research firm estimates vary widely and are treated as directional. Currency: INR; USD conversions at ₹88/USD [A].

Executive summary

Question	Answer
TAM	**Global:** ~$0.45B (2026) spent on cost visibility/optimization for GPU workloads on Kubernetes [S+A]. **India:** ~₹78-190 Cr/yr (~$9-22M) at this product's own price points [A], growing fast with India's GPU build-out.
SAM	**India, Kubernetes-based GPU fleets reachable with read-only, self-hosted, INR-priced software:** ~885 organizations, ~₹38 Cr/yr (~$4.4M) [A].
SOM (3 yrs)	**27-45 paying customers, ~₹0.8-1.35 Cr ARR (~$92-153k)** [A]. Viable as a bootstrapped business; not venture-scale on this product alone.
Main competitor	CAST AI ($1B valuation, Jan 2026) and ScaleOps ($800M+, Mar 2026) automate; Datadog launched GPU Monitoring (Apr 2026) and overlaps most directly on detection. You win only where read-only + self-hosted + INR + non-hyperscaler clusters matter.
Unit economics	Gross margin 78-89% (no hosting cost; you're self-hosted). LTV:CAC 3.4x (Starter) to 18x (Enterprise); CAC payback 4-6 months, founder-led [A]. Break-even ~13 customers at a ₹4L/month burn [A].
Biggest risk	**One-time-audit churn:** a customer finds the waste, fixes it, cancels. If churn doubles, Starter LTV:CAC drops to 1.7x (unprofitable). Also: GPU-count pricing underprices H100 fleets and overprices cheap-GPU fleets.

1. Market context

Signal	Figure	Tag
India end-user public cloud spending (Gartner)	$17.5B in 2026, up 28.1% from $13.7B in 2025	[S]
India data-center GPU market (MarkNtel)	~$0.17B (2025) to $5.29B (2032), 63% CAGR	[S]
IndiaAI Mission compute	38,000+ GPUs onboarded at ~₹65/GPU-hr; target 100,000 by Dec 2026	[S]
Indian GenAI startups (Nasscom)	890+ in H1 CY2025 (740+ application layer); $990M cumulative funding	[S]
India GPU installed base (third-party survey)	80,000+ GPUs across public/private deployments	[S, 3rd-party blog]

Signal	Figure	Tag
Global GPU-as-a-Service (MarketsandMarkets)	$8.21B (2025) to $26.62B (2030), 26.5% CAGR	[S]
Global Kubernetes cost management market	~$1.75B (2025) to ~$2.23B (2026), ~27% CAGR; other estimates $1.38B (2025)	[S, analyst range]
Kubernetes for AI inference (CNCF 2025 survey)	66% of orgs hosting GenAI use Kubernetes for some or all inference	[S]
GPU share of compute cost (Datadog)	GPU instances = 14% of compute costs across its customer base	[S]

What this means: the value pool (GPU spend) is growing 25-60% a year, and Kubernetes is now the default place GPU inference runs. But the software pool this product can capture is a thin slice of GPU spend: cost-visibility tools typically capture 1% or less of the spend they watch.

2. TAM, SAM, SOM

2.1 Global TAM (top-down, for context)

Global Kubernetes cost-management spend is ~$2.23B in 2026 [S]. We assume 15-25% of it is GPU-driven [A], anchored on Datadog's finding that GPUs are already 14% of compute cost and rising [S], and on GPU nodes costing several times more per hour than CPU nodes. That gives **~$0.33-0.56B, midpoint ~$0.45B (~₹3,900 Cr)** of global spend on cost visibility and optimization for GPU workloads on Kubernetes. This is the category CAST AI, ScaleOps, Kubecost and Datadog are fighting over. It is **not** reachable by a two-person, INR-priced, founder-led product in the next three years; it matters as the ceiling and as the reason incumbents will keep investing here.

2.2 India TAM (bottom-up, at this product's own prices)

Segment (orgs running dedicated NVIDIA GPU fleets, 2026)	Count	Basis
AI / GenAI startups that train or self-host models	~900	[A] from Nasscom's 890+ GenAI startups [S]: most of the ~150 infra/model-layer firms, a minority of the 740 app-layer firms (most call APIs), plus non-GenAI AI firms (vision, speech, health AI)
Enterprises with in-house GPU clusters (BFSI, telecom, IT services, e-commerce, auto)	~600	[A]
Research, academic and government labs (IITs, IISc, IndiaAI compute users)	~300	[A]
GPU clouds / neoclouds / MSPs (E2E, Yotta, Neysa, NxtGen, Netweb, smaller)	~30	[A] better treated as channel partners than end customers
Total	**~1,800**	

Tier (current pricing)	GPUs	Annual price	Assumed share of orgs [A]
Starter	1-5	₹1.2 L	45%
Growth	6-25	₹3.6 L	35%
Scale	26-100	₹9.0 L	15%
Enterprise (assumed ₹2 L/mo)	100+	₹24 L	5%
Blended ACV at list		**₹4.35 L**	

Method A (by organization): 1,800 orgs x ₹4.35 L = **~₹78 Cr/yr (~$8.9M)**. **Method B (by GPU):** 80,000 GPUs [S, third-party] x ~₹2,000 per GPU-month (the Growth tier's effective per-GPU price at ~15 GPUs) x 12 = **~₹192 Cr/yr (~$21.8M)**. India TAM is therefore **~₹78-190**

Cr/yr. Method B is higher because it counts GPU-cloud and IndiaAI capacity that sits in a few very large fleets. With IndiaAI alone targeting 100,000 GPUs by Dec 2026 [S], this number should roughly double by 2027-28.

2.3 SAM (India, serviceable by the product as built)

The product requires Kubernetes, NVIDIA GPUs with DCGM, and a buyer willing to install a Helm chart. Filters applied [A]:

- Exclude GPU clouds as direct customers (channel instead): 1,770 end-user orgs.
- Keep only fleets run on Kubernetes: 50% [A]. This is deliberately below CNCF's 66% [S] because many Indian startups run GPUs on bare VMs, Slurm, or notebooks rather than Kubernetes.
- Result: **~885 orgs x ₹4.35 L = ~₹38.5 Cr/yr (~$4.4M)**.

Not filtered out, but harder to win inside the SAM: teams on EKS/GKE/AKS that already have CAST AI, Datadog, or hyperscaler cost tools, and teams whose GPU spend is subsidized (IndiaAI at ~₹65/GPU-hr shrinks the rupee value of the same waste roughly 4-5x vs. commercial A100 rates).

2.4 SOM (3-year obtainable)

Scenario	Share of SAM	Paying customers	Realized ACV [A]	ARR
Conservative	3%	27	₹3.0 L	₹0.81 Cr (~$92k)
Base	4%	35	₹3.0 L	₹1.05 Cr (~$119k)
Stretch	5%	45	₹3.0 L	₹1.35 Cr (~$153k)

Realized ACV is below the ₹4.35 L list blend because early customers skew small and pilots get discounted [A]. 3-5% of SAM in three years is normal for a founder-led product with no sales team.

Honest read: the product as built supports a profitable small business (break-even is ~13 customers, section 4), not a venture-scale one. Getting to venture scale needs one or more of: (1) selling outside India (SE Asia, Middle East, and self-managed/neocloud fleets globally, a ~$20M+ SAM if 5% of the global TAM is reachable [A]); (2) white-labelling through Indian GPU clouds to reach their customers; (3) moving from detection to action (the queue-depth/TTFT and bin-packing items from the MVP roadmap), which is what justifies CAST AI's and ScaleOps's valuations.

3. Competitive landscape

Four kinds of competitor: **automation platforms** that change your cluster (CAST AI, ScaleOps, PerfectScale), **cost-allocation tools** (Kubecost/OpenCost), **observability platforms** adding GPU views (Datadog), and **schedulers** (NVIDIA Run:ai / KAI). The default competitor is still **DIY**: dcgm-exporter + Prometheus + Grafana.

Competitor	What it is	GPU capability	Pricing (public / reported)	Scale	Overlap
CAST AI	Kubernetes automation platform (autoscaling, bin-packing, Spot, rightsizing)	GPU node autoscaling, Spot GPUs, time-slicing/MIG sharing, GPU cost monitoring, OMNI Compute / GPU marketplace	Free monitoring; paid tiers reported at $200 / $1,000 / $5,000 per month; savings-share ~15-20% reported (third-party)	$1B valuation (Jan 2026); ~$181M raised	High on visibility; they act, you recommend
ScaleOps	Autonomous real-time Kubernetes resource management	Adjusts compute and GPU resources automatically	Not public (enterprise)	$130M Series C at $800M+ (Mar 2026); $210M total; 350%+ YoY growth	Medium-high
Datadog GPU Monitoring	Observability platform add-on (GA Apr 22, 2026)	Fleet health, idle/misconfigured GPUs, cost + chargeback by team, cloud / neocloud / on-prem	Usage-based on top of Datadog agent pricing	Public company; huge installed base	**Highest on detection**
IBM Kubecost / OpenCost	Kubernetes cost allocation (acquired by IBM Sep 2024; OpenCost is the CNCF open-source core)	Per-GPU hourly cost allocation; idle-capacity cost; not GPU-utilization waste classification	Free/OSS tier; Business plan reported ~$449/mo per 100 nodes	IBM FinOps suite (Apptio, Cloudability, Turbonomic)	Medium
PerfectScale (DoiT)	Kubernetes optimization and governance (acquired by DoiT Feb 2025)	Rightsizing across CPU, GPU, storage, networking	Bundled with DoiT services	Part of DoiT	Medium
NVIDIA Run:ai / KAI Scheduler	GPU scheduler/orchestrator; KAI is the Apache 2.0 open-source core (CNCF Sandbox)	Fair-share quotas, gang scheduling, GPU sharing; utilization dashboards in Run:ai	KAI free; Run:ai enterprise	NVIDIA	Low (different layer)
DIY: DCGM + Prometheus + Grafana	Free open-source metrics stack	Raw utilization per GPU; no waste taxonomy, no workload rollups, no ₹ recommendations out of the box	Free (engineer time)	Everywhere	High on raw data
Indian GPU clouds' own consoles	E2E, Yotta, etc. instance dashboards	Basic per-instance utilization for their own GPUs	Included	Growing fast	Low; potential channel partners

3.1 Feature comparison vs. CAST AI (closest well-known name)

Capability	Your product	CAST AI
GPU waste detection (idle / over-provisioned / fragmented / recurring)	Yes, four categories, confidence-scored	Partial (utilization + cost monitoring; not this taxonomy)
₹ savings estimate per finding	Yes, native INR, SLA-aware right-sizing ceiling	USD cost; savings reported after automation
Automatic action (scale, rebin, Spot, sharing)	**No** (recommends only)	**Yes** (core product)
Cluster permissions	**Read-only** (get/list/watch only)	Needs write access for automation
Where data lives	**Customer's own cluster** (self-hosted, SQLite)	CAST AI SaaS
Cluster types	Any Kubernetes with NVIDIA+DCGM; tested GKE and k3s/vast.ai	Strongest on EKS/GKE/AKS
Cloud-billing integration	No	Yes
Serving metrics (queue depth, TTFT)	No (roadmap)	Partial
Multi-cluster, SSO, SOC 2	No	Yes
Pricing model	Flat INR subscription by GPU count	USD tiers + savings share

3.2 Where you win and where you lose

You win when...	You lose when...
- Security won't grant write access to a third-party agent (banks, government, regulated health) - Data must stay in-cluster or in-country (DPDP Act, sovereign-AI projects) - The cluster is on-prem, a neocloud, or k3s, not EKS/GKE/AKS - The buyer wants a flat INR invoice, not a USD savings share - The team is small (5-50 GPUs) and not a Datadog customer	- The buyer wants savings to happen automatically (CAST AI, ScaleOps) - They already pay Datadog and can switch on GPU Monitoring - They need billing reconciliation, multi-cluster, SSO, SOC 2 - Their GPUs are subsidized and the waste isn't worth ₹ to them - An engineer says "we'll build it in Grafana"

Defensibility is low today. The detection logic can be copied, and Datadog shipped a similar capability this year. Your advantages are distribution (India, INR, founder-led), the trust model (read-only, self-hosted), and speed. The durable moat has to come later from real pilot data (calibrated thresholds, per-workload baselines) and from being the tool Indian GPU clouds bundle.

4. Unit economics

The product is self-hosted in the customer's cluster, so you pay nothing for compute or storage per customer. Cost of goods is mostly **your support time**. All inputs below are assumptions [A] until pilots produce real data; founder time is costed at ₹1,500/hour [A].

4.1 Per-customer, monthly

Line item	Starter	Growth	Scale	Enterprise
Price / month	₹9,999	₹29,999	₹74,999	₹2,00,000 [A]
Hosting / compute	₹0	₹0	₹0	₹0
Shared tooling (CI, registry, email, domain)	₹500	₹500	₹500	₹500
Support hours / month [A]	1 h = ₹1,500	2 h = ₹3,000	5 h = ₹7,500	12 h = ₹18,000
Payment processing (2%) [A]	₹200	₹600	₹1,500	₹4,000
COGS	₹2,200	₹4,100	₹9,500	₹22,500
Gross profit / month	**₹7,799 (78%)**	**₹25,899 (86%)**	**₹65,499 (87%)**	**₹1,77,500 (89%)**

4.2 Acquisition, lifetime value, payback

Metric	Starter	Growth	Scale	Enterprise
CAC: founder-led sales, incl. ~4 free scans per win [A]	₹40,000	₹90,000	₹2,50,000	₹6,00,000
Onboarding (install, 7-day scan, readout) [A]	4 h = ₹6,000	8 h = ₹12,000	16 h = ₹24,000	40 h = ₹60,000
Total acquisition cost	₹46,000	₹1,02,000	₹2,74,000	₹6,60,000
Monthly churn [A]	5%	3%	2%	1.5%
Expected lifetime	20 months	33 months	50 months	67 months
LTV (gross profit x lifetime)	₹1.56 L	₹8.63 L	₹32.7 L	₹1.18 Cr
LTV : CAC	**3.4x**	**8.5x**	**12.0x**	**17.9x**
CAC payback	5.9 months	3.9 months	4.2 months	3.7 months
LTV : CAC if churn doubles	1.7x	4.2x	6.0x	9.0x

Rule of thumb: LTV:CAC above 3x and payback under 12 months is healthy. Every tier clears that on paper, but **Starter is marginal**: it breaks if churn is higher than assumed, and it breaks as soon as a paid salesperson replaces founder time. CAC is founder-time CAC; hiring an account executive at ~₹1.5-2 L/month [A] roughly doubles to triples CAC for the lower tiers.

4.3 Customer-side ROI: does the price make sense to the buyer?

Recoverable waste assumed at **10% of GPU spend** [A]. Basis: the Philly replay flagged 23.0% of GPU-workload-hours as waste [S, own test]; not all flagged waste is recoverable (SLA headroom, team inertia), so we assume 30-50% of it gets acted on, i.e. ~7-12%. GPU rates are the product's own illustrative INR defaults (L4 ₹70/h, A100 80GB ₹300/h, H100 SXM ₹520/h), and IndiaAI's ~₹65/h [S]; 730 hours/month.

Customer fleet	GPU spend / month	Tier	Price as % of spend	10% recovered	ROI (x price)
4 x L4	₹2.04 L	Starter	4.9%	₹20,440	**2.0x** (weak)
16 x GPUs on IndiaAI subsidy	₹7.59 L	Growth	4.0%	₹75,920	**2.5x** (weak)
4 x A100 80GB	₹8.76 L	Starter	1.1%	₹87,600	**8.8x**
16 x A100 80GB	₹35.0 L	Growth	0.86%	₹3.50 L	**11.7x**
64 x H100 SXM	₹2.43 Cr	Scale	0.31%	₹24.3 L	**32x** (underpriced)

Pricing problem: pricing by GPU count ignores that an H100 costs ~7x an L4. Small fleets of cheap or subsidized GPUs get a weak 2-2.5x return (a 2x ROI is usually not enough for a buyer to bother), while large H100 fleets pay 0.3% of spend for a 32x return. **Recommendation:** price on monitored GPU spend, e.g. ~1% of monitored monthly GPU spend with a ₹9,999 floor and volume discounts [A]. That keeps ROI around 10x for everyone and grows revenue with the customer's fleet.

4.4 Company break-even

Assumed monthly burn of **₹4 L** [A] (two founders at ₹1.5 L each, ~₹30k GPU test rentals, ~₹40k legal/accounting/tools). At the assumed tier mix (45/35/15/5), the average customer contributes ~₹31,300/month gross profit, so **break-even is ~13 paying customers** (or ~16 Growth-only, or ~52 Starter-only). That is well inside the 3-year SOM of 27-45 customers.

4.5 What would change these numbers most

- **Churn** (the biggest swing factor). If customers treat the product as a one-time audit, lifetime collapses. Counter-measures: continuous value (weekly ₹ reports, regression alerts when waste returns, per-team chargeback) and annual contracts.
- **Free-scan conversion.** CAC assumes ~1 in 4 free scans converts. At 1 in 10, lower-tier CAC more than doubles.
- **Pilot-proven savings.** The whole ROI table rests on the 10% recoverable assumption. The first pilot's before/after number replaces it.
- **Moving to action** (bin-packing recommendations, queue-depth-aware right-sizing) supports a higher price and lower churn, but brings you into direct competition with CAST AI and ScaleOps.

5. Assumptions ledger

#	Assumption	Value	Why / how to validate
A1	USD/INR	₹88	Approximate 2025-26 rate
A2	GPU share of Kubernetes cost-management spend	15-25%	Datadog: GPUs 14% of compute cost and rising; GPU nodes cost more per hour
A3	Indian orgs running dedicated NVIDIA GPU fleets	~1,800	Built from Nasscom startup counts + estimates; validate via GPU-cloud partner conversations
A4	Share of those on Kubernetes	50%	Below CNCF's 66%; ask every prospect
A5	Tier mix of customers	45/35/15/5	Skewed small; replace with pipeline data
A6	Enterprise price	₹2 L/month	Custom tier; placeholder
A7	Realized ACV	₹3.0 L	Discounts + small-customer skew
A8	Founder time value	₹1,500/hour	Opportunity cost placeholder
A9	Support / onboarding hours	See 4.1-4.2	Measure on first pilots
A10	CAC per tier	₹40k to ₹6 L	Founder-led, ~4 free scans per win
A11	Monthly churn	1.5-5%	Biggest uncertainty; see 4.5
A12	Recoverable waste	10% of GPU spend	From Philly 23% flagged x 30-50% acted on; prove in pilot
A13	Monthly burn	₹4 L	Two founders, lean
A14	3-year SAM capture	3-5%	Typical for founder-led B2B infrastructure software

6. Sources

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- Own test results: testing/real-trace-replay/RESULTS.md (Philly, 23.0% flagged); testing/azure-inference-trace-replay/RESULTS.md; PROGRESS.md.

Market-research-firm figures differ substantially between publishers; treat them as ranges. Competitor pricing marked third-party was not confirmed on the vendor's own pricing page.